



CSD 3183

Pokemon Trainer Move Selection
AI (turn-based)

Presented by
Team 2



Project Description

Core Decision-Making Architecture

- Pokémon is a turn-based game in which both players choose their actions simultaneously, without knowing the opponent's actions
- Classical adversarial algorithms require the turns to be strictly alternating, making it unsuitable for the game
- Greedy agent, Simultaneous Move Alpha-Beta (SMAB) agent, Information Set Monte Carlo Tree Search (IS-MCTS) agent will be explored and built to compare and address the problem





Project Relevance

01

Breaks alternating turn assumption used in traditional tree-searching algorithms

02

Requires statistical optimisations instead of brute forcing

03

Can be applied to many relevant real-world challenges

Pikachu Lvl 49

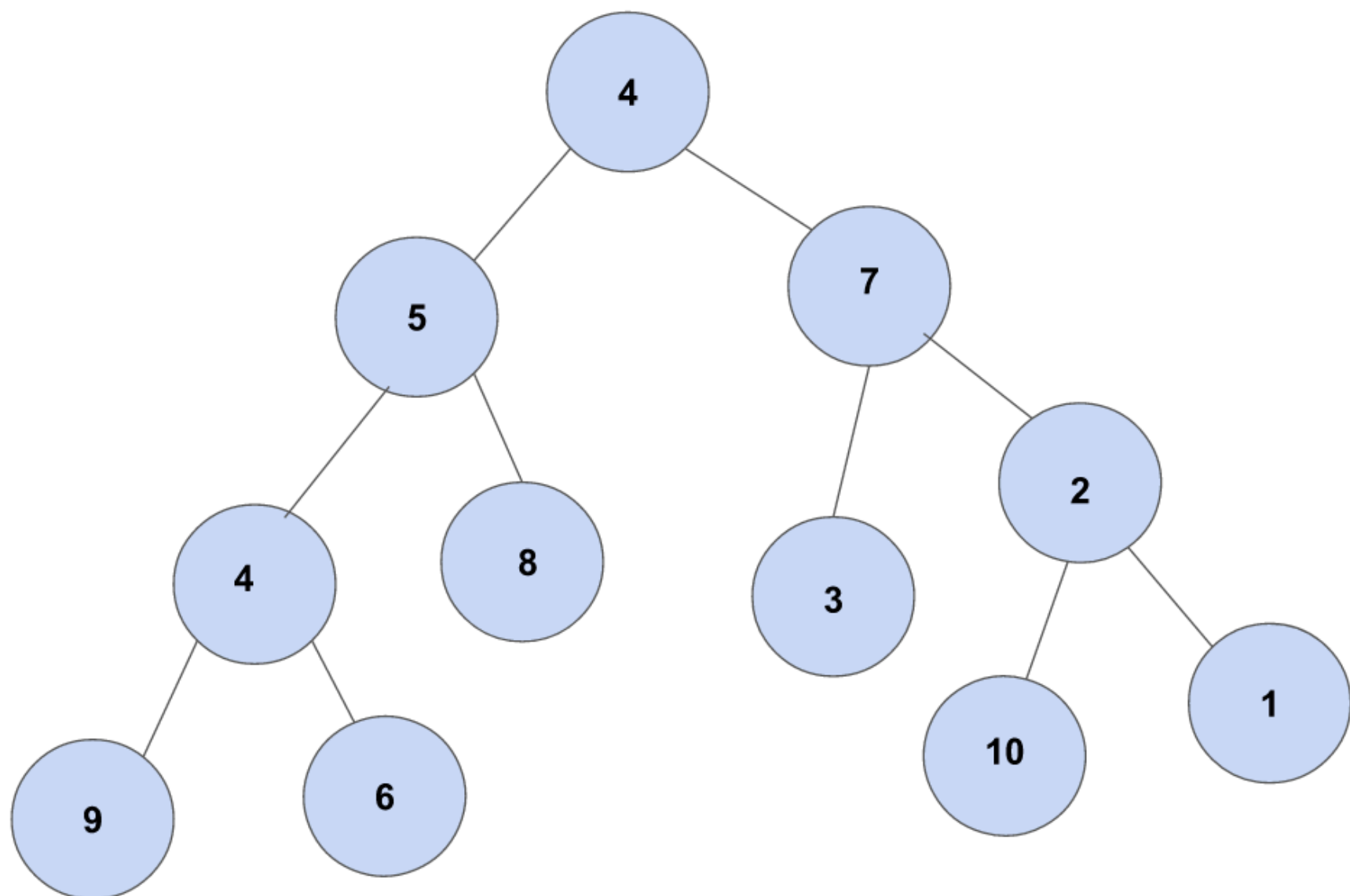


Charizard Lvl 45
HP: 136/136



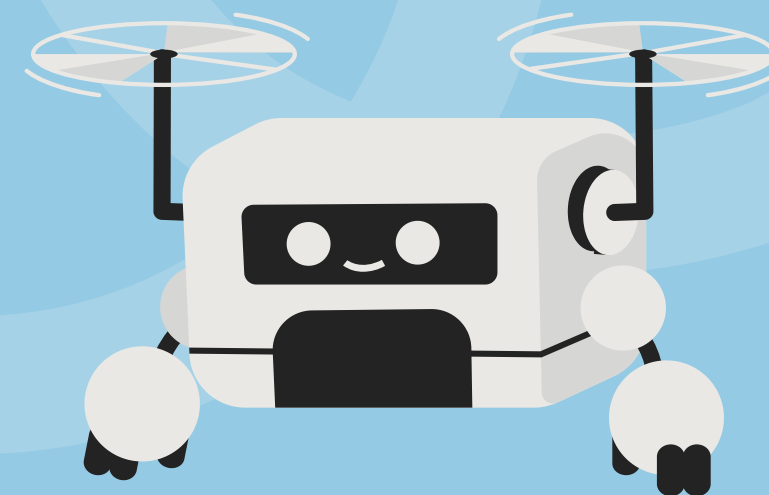
Trainer Blue challenges you!
Go! Charizard!





Greedy Agent

It finds an optimal solution (best immediate reward) after every turn, building the solution along the way.



- Fast and simple to implement

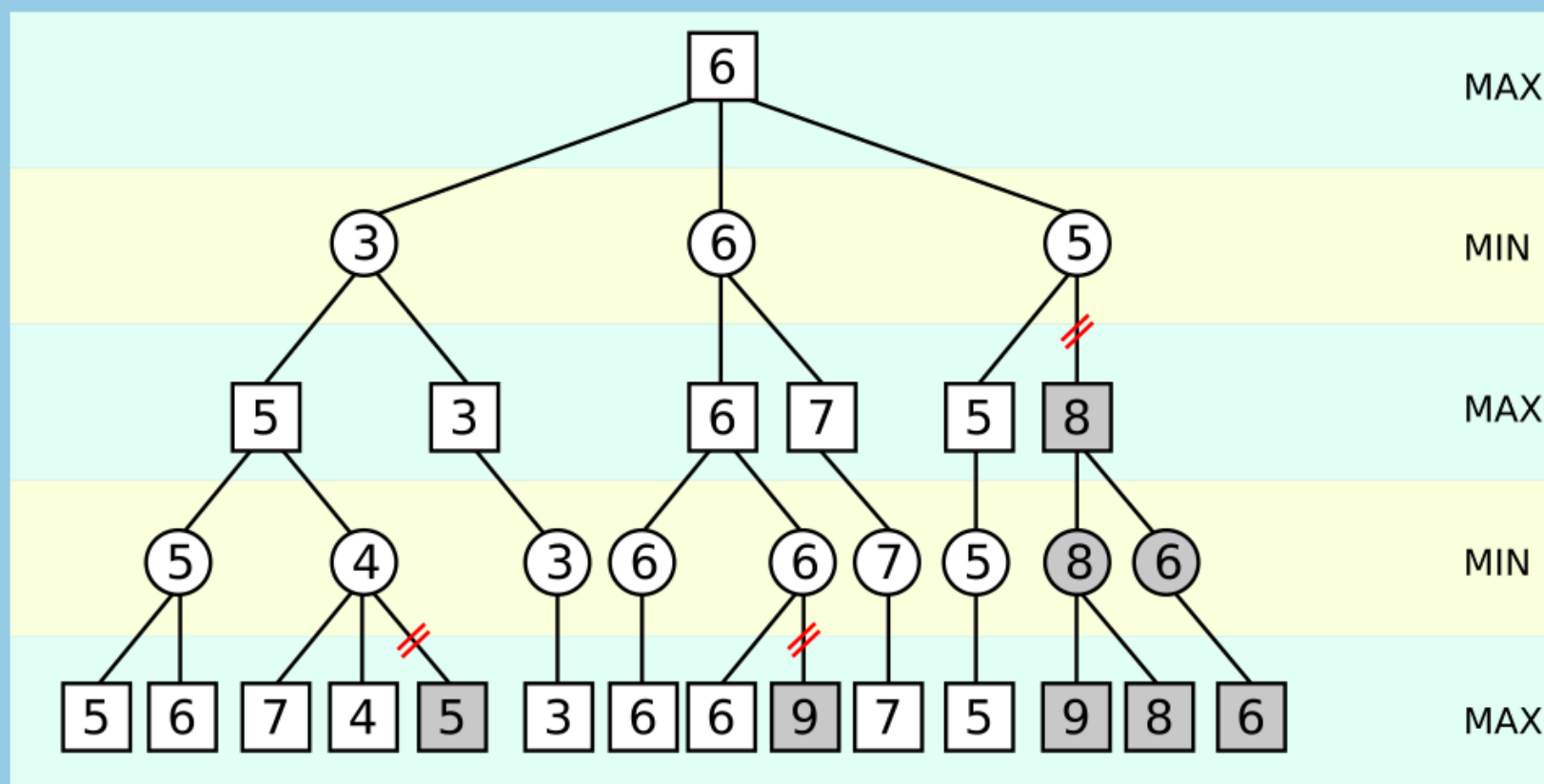
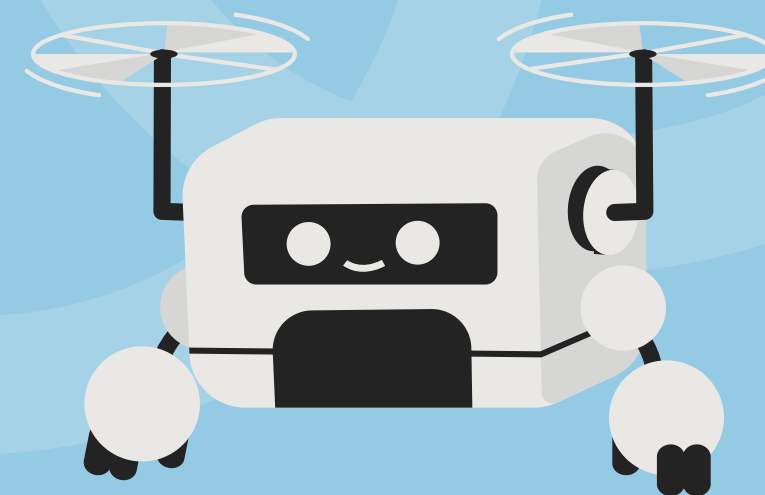


- Only considers the immediate conditions, bad for long term





SMAB Agent



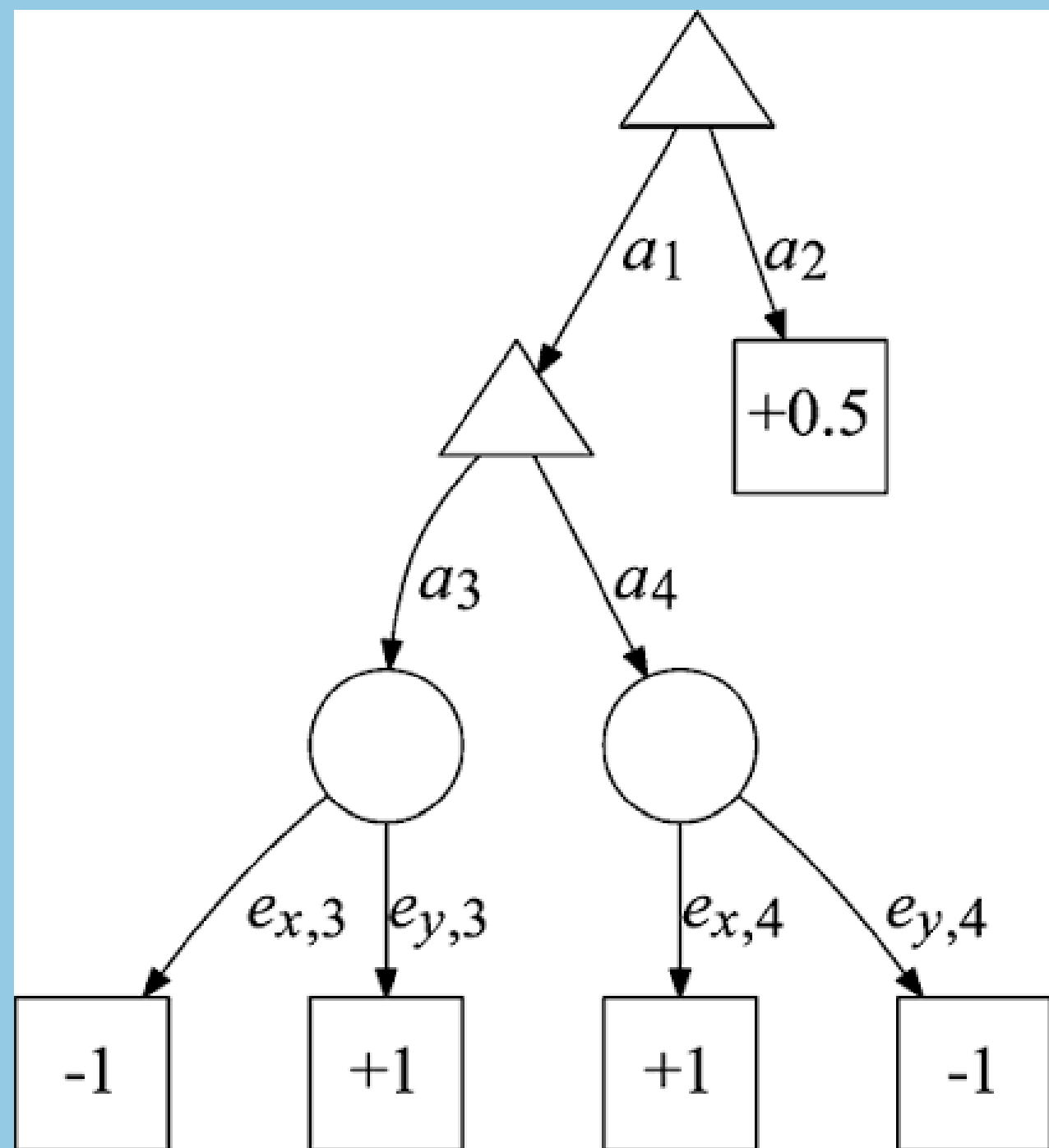
It models each decision point as a zero-sum payoff matrix and searches for the action that maximises the player's worst-case outcome; in this case, it is receiving serious damage to their own Pokémon.

- Considers possibilities far into the future, making optimal choices while pruning bad choices

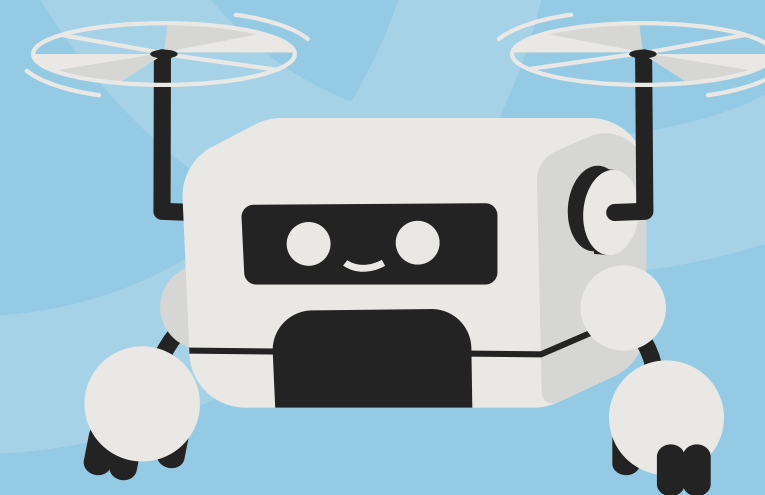


- Considers everything, so more variables and more players make it less effective and expensive





IS-MCTS Agent



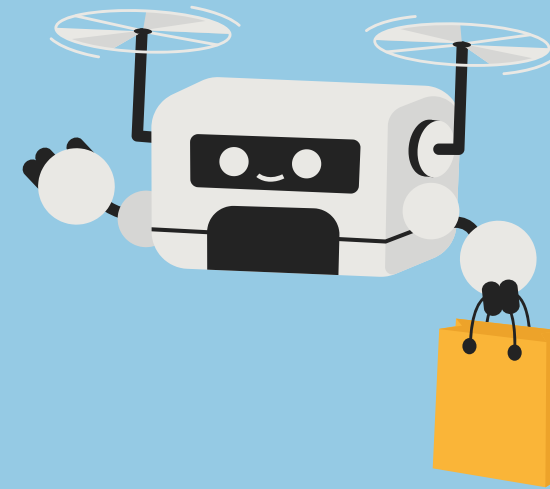
It randomly simulates game scenarios based on the current information it has and estimates which action will likely allow the player to win via the simulated results. By doing so, it is more efficient than checking every possible game scenario, which would be too slow.

- Able to approximate the best choice without extensive searching



- Success rate depends on size of approximations





Benchmark Comparison

- Set minimum Pokémon level to 60
- Set maximum Pokémon level to 61
- 500 matches

01

Greedy Agent

```
=== BENCHMARK RESULTS (Greedy vs Greedy) ===  
Total Matches: 500  
Greedy Wins: 245 (49%)  
Greedy Wins: 255 (51%)  
Total Simulation Time: 243.29 ms  
Average Time Per Match: 0.486581 ms  
=====
```

02

SMAB Agent

```
=== BENCHMARK RESULTS (SMAB vs Greedy) ==  
Total Matches: 500  
SMAB Wins: 359 (71.8%)  
Greedy Wins: 141 (28.2%)  
Draws: 0  
Total Simulation Time: 146460 ms  
Average Time Per Match: 292.919 ms  
=====
```

03

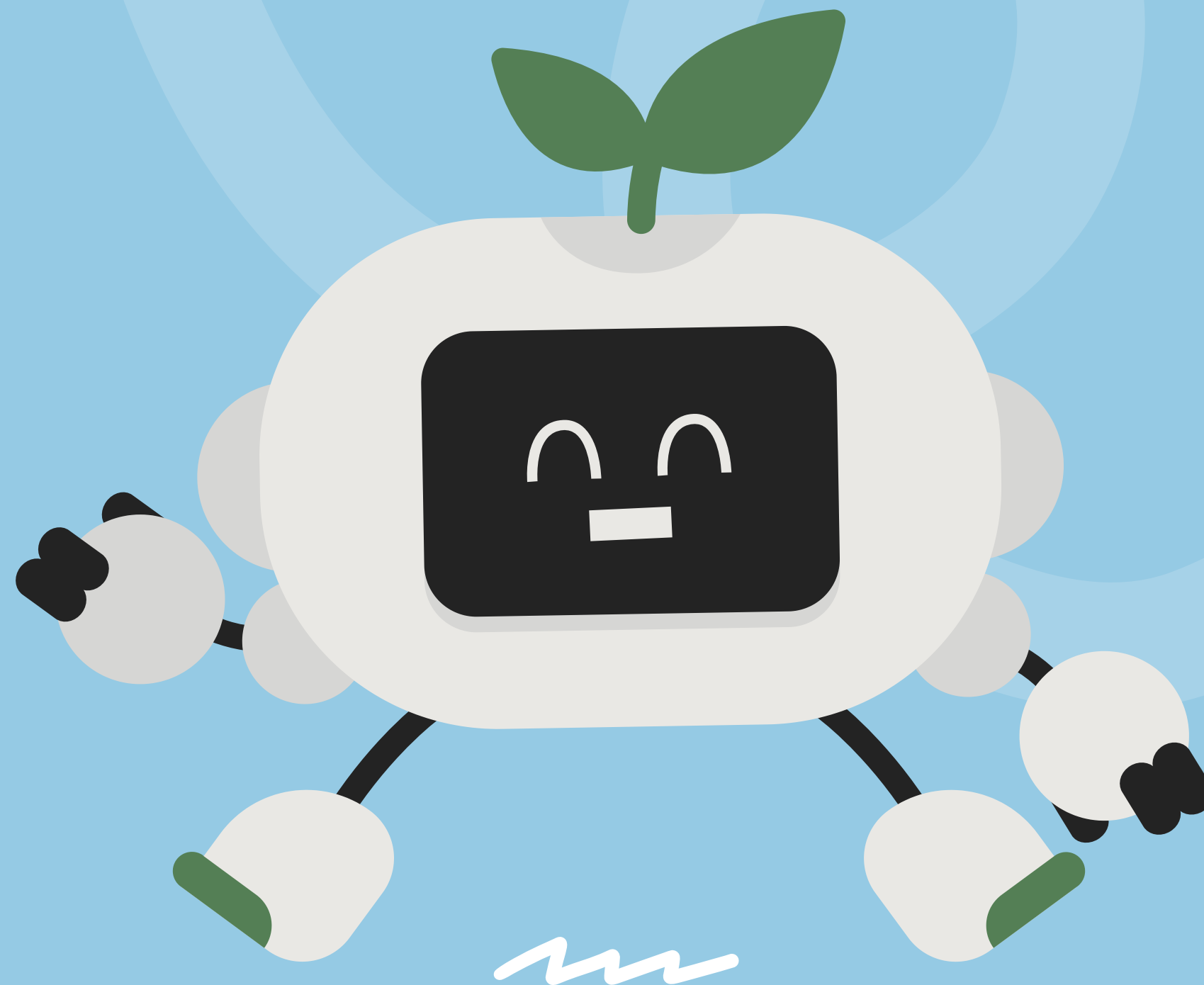
IS-MCTS Agent

```
=== BENCHMARK RESULTS (MCTS vs Greedy) ==  
Total Matches: 500  
MCTS Wins: 345 (69%)  
Greedy Wins: 155 (31%)  
Draws: 0  
Total Simulation Time: 524759 ms  
Average Time Per Match: 1049.52 ms  
=====
```





Demo





Thank You!

Q&A Time

